

Improving Provenance Data Interaction for Visual Storytelling in Medical Imaging Data Exploration

submitted to EuroVis 2018

Comments and replies

We thank the anonymous referees for their reports and comments. We have carefully addressed all the issues raised and prepared a revision of our manuscript. Here is a summary of our response to the reports and the changes we have made.

Coordinator review:

The primary weakness is the lack of explanation of the current state of radiology analysis software and what drove this software to be produced. Do current software systems lack the storytelling?

Reply: According to our respondents from the preliminary survey and the literature, the current radiology analysis software focuses on image data (CT, MRI) analysis. During analysis and decision making, several outputs are generated, such as audio recording during exploration, key images, and a detailed report. Although all these pieces of information are related to each other in the diagnosis process, they exist separately. They lack the storytelling aspect both for exploration and presentation purposes. Our work ties this information into a visual story and thus creates a connection between the images, exploration, findings and conclusions to be drawn from the data. We have made this clearer in the introduction.

I had a difficult time understanding figures 3 and 4 and spent some time going back to the text, I am not sure how to improve on that in the 4 page limit. Perhaps figure 3 could be halved to make room for the comparison against existing methodologies?

In Figure 3 my problem started because the caption notes there are four levels of abstraction, but only three are labeled in the image. It took me a while to figure out the first one was the whole thing... The right side is perhaps redundant and space consumptive.

Reply: We removed one diagram, and also improved Figure 3. Now the figure clearly shows how data would be visualized at four different levels of abstraction. However, it is difficult to make a comparison against existing methodologies since through the preliminary survey and to our knowledge, visual storytelling concept has not been integrated into current radiology analysis software (yet). [GLG 16] presents a storytelling tool based on the Phovea framework, however they did not employ the grouping strategy as we show in Figures 3 and 4.*

Figure 1. All of the text needs to be a bit larger and thus legible.

Reply: We have increased the font size of the text and its layout to make it more readable.

Committee member review #1:

It is unclear to me how user's intent is captured here, since those interaction are not mutual exclusive or independent. Take Figure 4 as an example, it is unclear if the user's rotation is for explore or for select. Thus the node aggregation

seems to be somewhat arbitrary.

The node aggregation is based on a defined categorization, as we explained in Section 4.2. At this stage, we simply categorize each user interaction (each exploration step) using an adjusted version of the taxonomy made by Yi et al. However, we see the point raised and we agree that more sophisticated (and possibly exhaustive) rules are required in the grouping based on user's intention. We have revised Section 6 and made it clearer that we will further investigate the taxonomy.

The paper is easy to understand and well written. Please use the third-person to refer to yourself, when citing your own work. Otherwise, the paper is not anonymous, unfair to other paper authors.

Reply: Although we did not cite our own work in the submission, we checked all the citations again to avoid unnecessary misunderstanding.

Committee member review #2:

My big issue with this paper is that the requirements and guidelines made by the authors are very vague. Although this is a short paper I would at least expect one evaluation setting with a user group from the medical domain or at least user statements if the presented approach fits their needs.

Reply: We agree with this and, for this reason, we aim to evaluate our tool and its features in the near future, as we now mention in the last section. Nevertheless, we have added some user statements in the Discussion section. Overall, the benefits of using our prototype and of creating visual stories over text reports is well-perceived.

As the paper states in the title, the system is especially designed for medical purposes. Besides some minor statements I don't see how this system differs from setting in other domains.

Reply: Although our visual storytelling tool and its features are generic and they can be adapted to different fields of application, it has been designed based on our survey and user needs in medical imaging data exploration: diagnostics in radiology is our primary use case. We have now made this clearer in the introduction. Then, assuming that the type of (image) data, type and size of provenance data, user intents, and interaction with provenance data differ from domain to domain, we believe that systems should differ as well. This is an interesting avenue for further research.

I am also missing a concrete example on how the system is used in a specific task (surgery or treatment planning would be nice here).

Reply: Thank you for your suggestion. To make it more understandable, we have added a concrete example to the introduction.